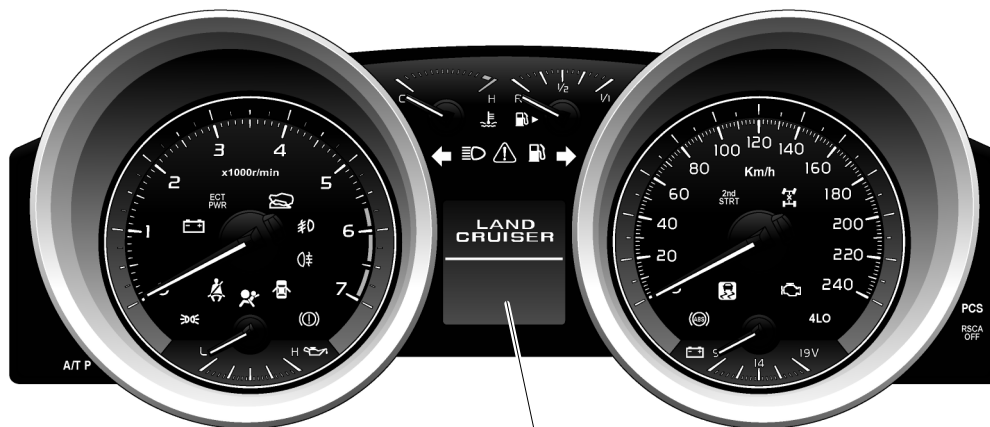


METER

■ COMBINATION METER

1. General

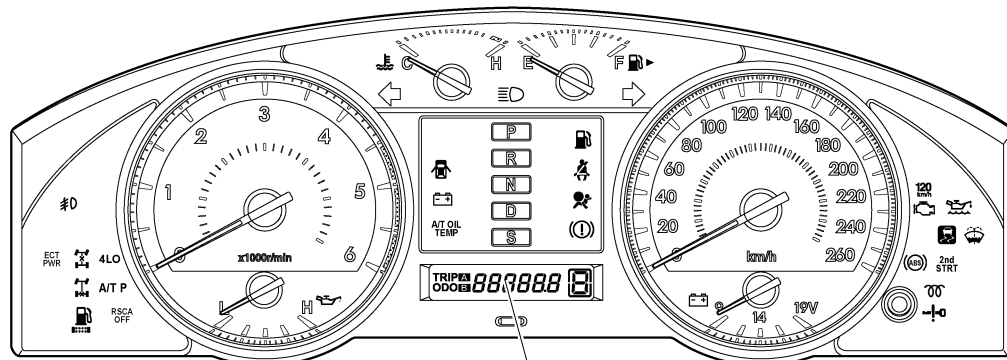
- An optitron display type combination meter is provided on VX grade models. The optitron display type meter realizes excellent visibility through the use of smoke acrylic in the protective panel, and bright LEDs (Light Emitting Diodes) that have high contrast to illuminate the dial.
- A multi-information display is mounted in the center of the optitron display type combination meter.
- An analog type combination meter is provided on GX grade models.
- A meter ECU and buzzer are enclosed in the combination meter. This ECU maintains communication with other ECUs through the CAN.
- Illumination control, which turns on the combination meter illumination at different time intervals when the power source is switched to IG-ON, has been provided.
- A speed warning system has been provided as standard on models for G. C. C. Countries. When the vehicle speed exceeds 120km/h, the system warns the driver by sounding the buzzer and flashing the warning lights.
- On dual fuel tank models, the meter ECU determines the remaining fuel amount in the fuel tank from the signals received from the sender gauges on both the fuel tank and fuel sub tanks.



Multi-information Display

European models with 2UZ-FE engine

081BE03Y



LCD (Liquid Crystal Display)

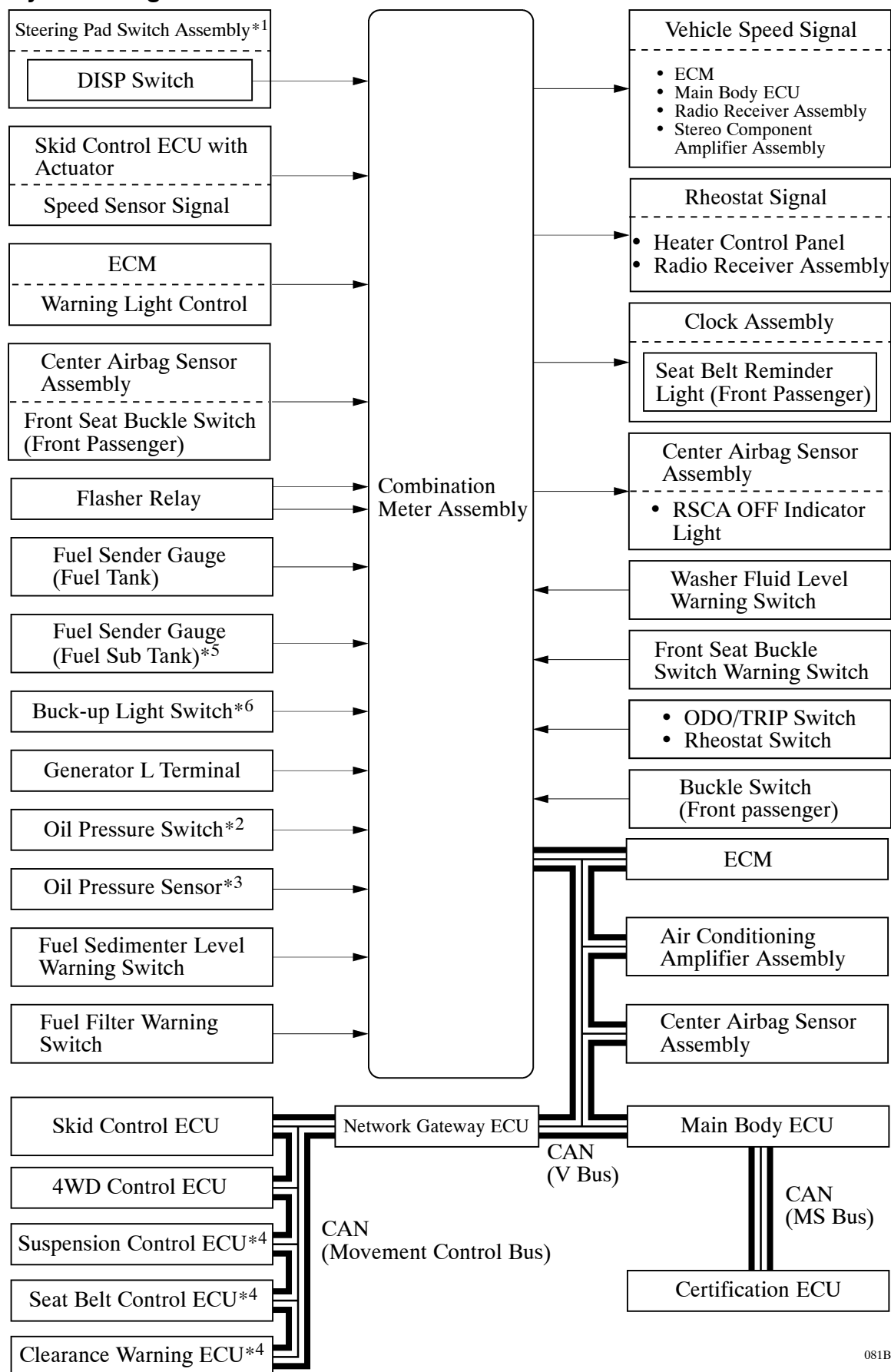
G.C.C. Countries models with 1VD-FTV engine

081BE04Y

Service Tip

If the LEDs malfunction, the entire combination meter assembly must be replaced. For details, refer to the Land Cruiser (Station Wagon) Repair Manual (Pub. No. RM0810E).

2. System Diagram



081BE47S

*1: Only for models with Optitron type Combination Meter

*3: Except 1GR-FE Engine

*5: Only for diesel engine models with dual tank

*2: Only for models with 1GR-FE Engine

*4: Optional Equipment

*6: Models with manual transmission

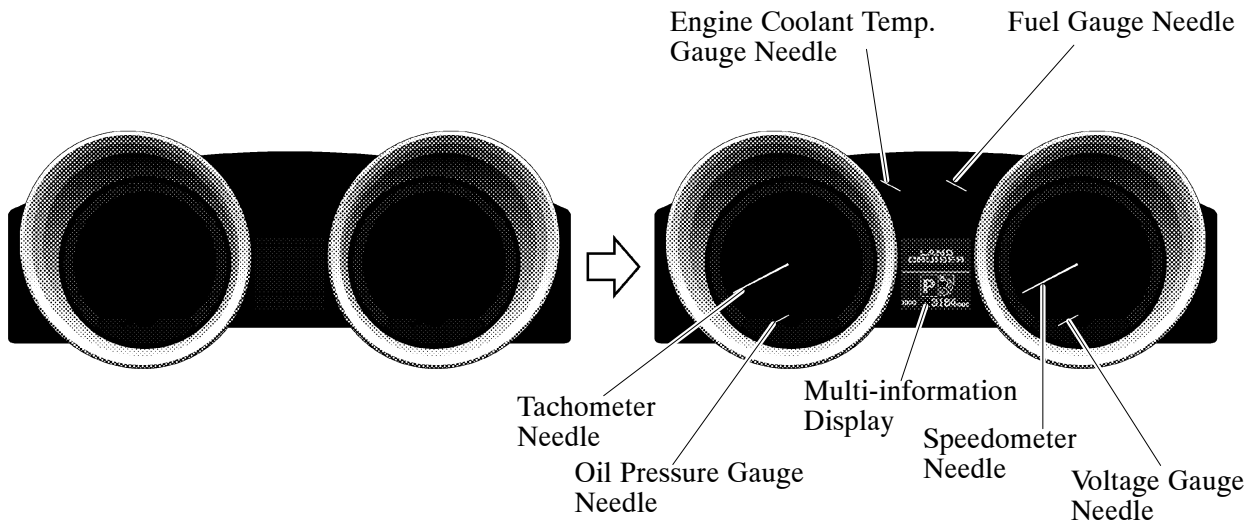
► Input and output communication signals of the combination meter ◀

ECU	Input Signal	Output Signal
ECM	• Engine speed • Shift position • S mode position • Fuel injection volume • Warning Light Control • Engine coolant temperature • Current range position • Buzzer sounding request • Indicator light illumination request • Warning message display request • Diagnosis (Cruise)	• Vehicle Speed • Remaining Fuel Amount*¹ (Fuel tank/ Fuel sub tank)
Air Conditioning Amplifier Assembly	• Diagnosis	Vehicle speed
Center Airbag Sensor Assembly	• Warning light illumination request • Seat belt remainder control (Driver and Front Passenger) • Diagnosis	Vehicle speed
Main Body ECU	• Lighting status • Parking brake switch • Courtesy switch • Warning message display request • Buzzer sounding request • Auto dimmer signal • Diagnosis	—
Skid Control ECU	• Warning message display request • Warning light illumination request • Indicator light illumination request • Diagnosis	—
Seat Belt Control ECU* ²	• Warning message display request • Warning light illumination request • Buzzer sounding request • Diagnosis	—
Clearance Warning ECU* ²	• Warning message display request • Buzzer sounding request • Intuitive parking assist status indication	Vehicle speed
Certification ECU	• Warning message display request • Buzzer sounding request	—
4WD Control ECU	• Indicator light illumination request • Buzzer sounding request	—
AHC ECU* ²	• Warning message display request • Warning light illumination request • Buzzer sounding request • Diagnosis	—

*¹: Only for diesel engine models with dual tank*²: Optional equipment

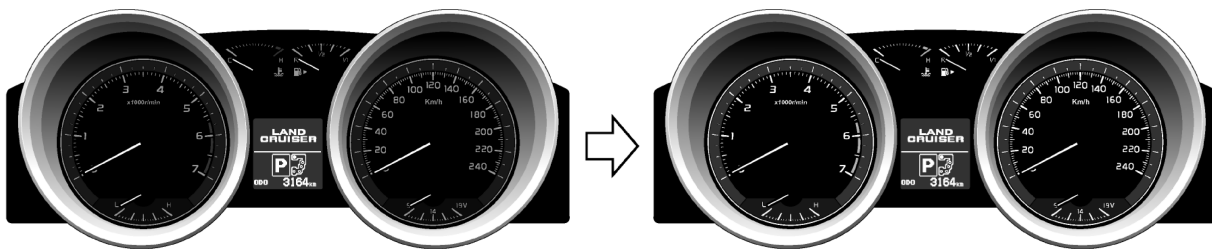
3. Illumination Control

- When the power source is switched to IG-ON, the illumination control operates as follows (1, 2). However, all illuminations turn off while the engine is cranked.
 - When the power source is switched to OFF, the illumination control is as follows (3).
- 1) The needles of the speedometer, tachometer, fuel gauge, engine coolant temperature gauge, and voltage gauge and oil pressure gauge are illuminated and “LAND CRUISER” appears on the multi-information display.



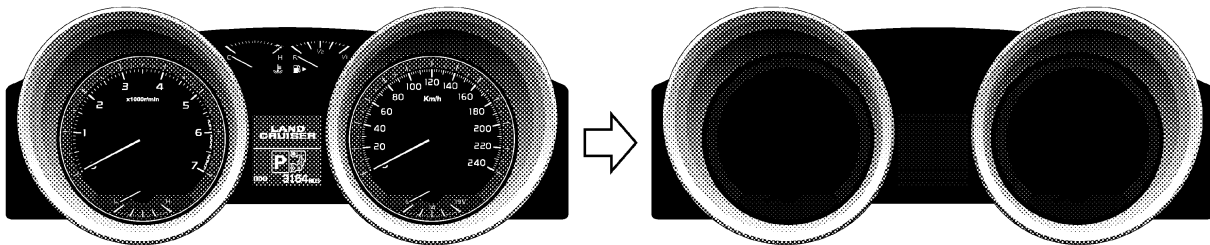
080BE09Y

- 2) After the multi-information display illuminates, the meter illumination gradually fades in.



081BE06Y

- 3) All illuminations fade out.



081BE07Y

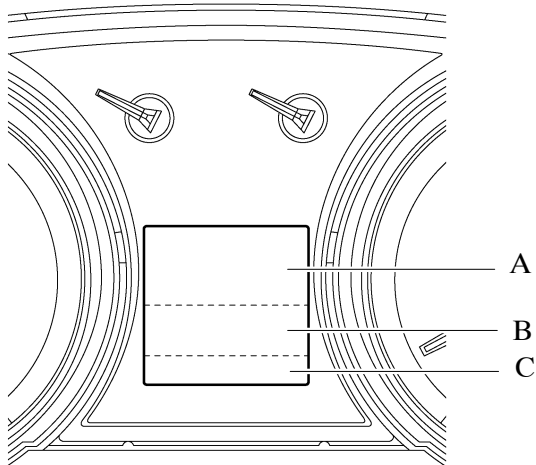
- ◆: The above illustration expresses only the transition of the illumination, therefore, the actual illumination transition is different from the illustration above.

4. Multi-information Display

General

The new Land Cruiser (Station Wagon) multi-information display has three areas depending on the display mode.

► Display Areas ◀



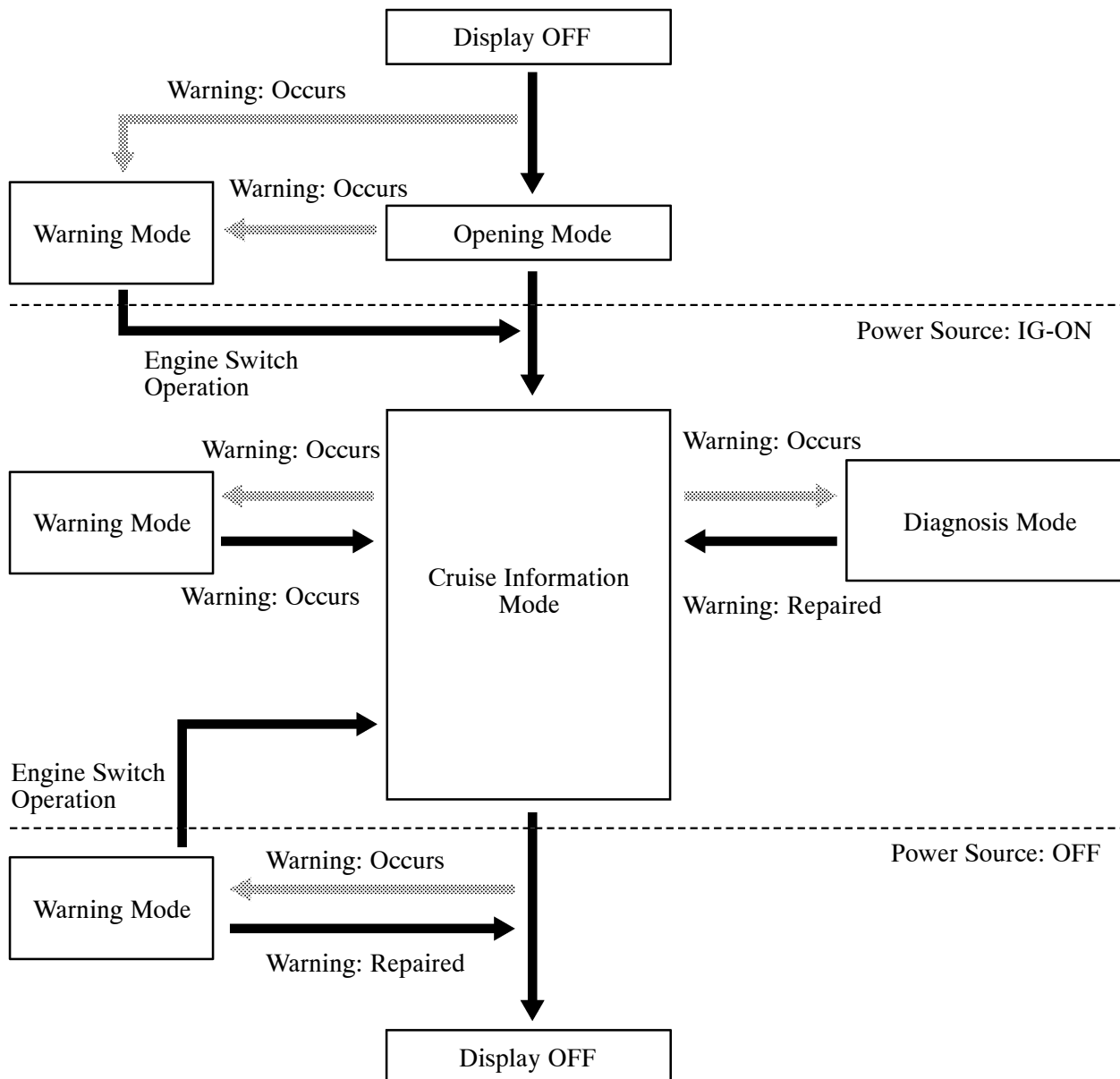
080BE73H

► Displayed Information ◀

Display Areas	Mode	Outline
A	Cruise Information (See page BE-45)	• Eight types of information (instant fuel consumption, average fuel consumption after refueling, driving range, average fuel consumption, average speed since engine start, distance driven since engine start) can be displayed. • The display can be switched by using the DISP switch.
	Warning (See page BE-46)	Interrupts the multi-information display immediately when a warning occurs.
	Clearance Sonar (See page BE-112)	Displays the location of an obstacle and the approximate distance between the vehicle and the obstacle.
	Diagnosis (See page CH-230, 248)	DTC (Diagnostic Trouble Code) for the brake control system (TRAC and VSC function), tire pressure warning system and VGRS can be displayed.
B	Shift Position	Displays the current shift position.
	Active Height Control Suspension	Displays the currently selected vehicle height.
C	ODO/TRIP	Switches the display of the ODO/TRIP and displays the ODO/TRIP.

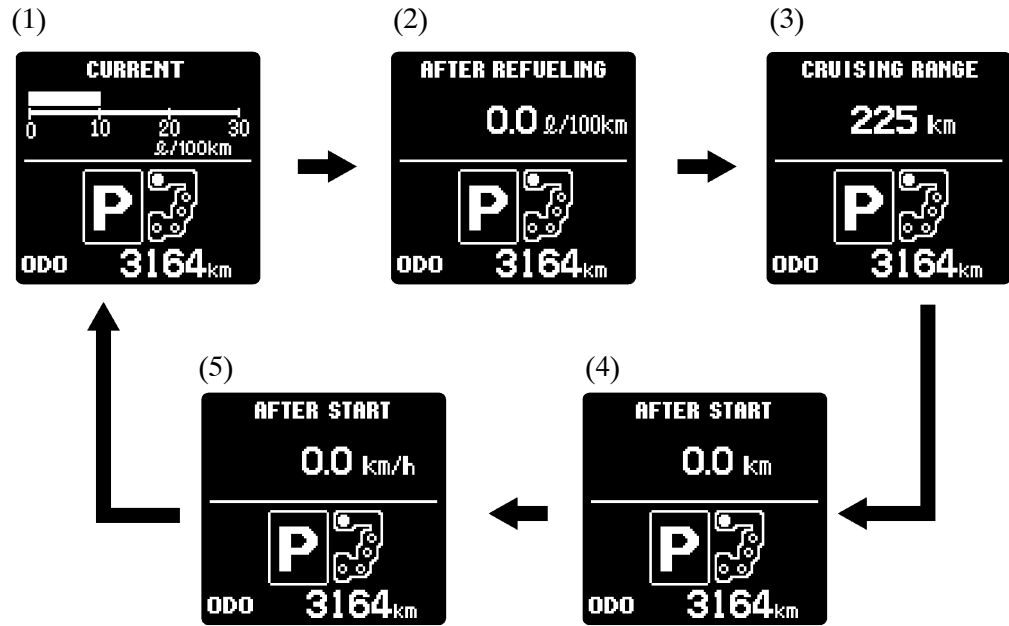
Flow of The Multi-information Display Indication

The display mode shown in area A on the multi-information display can be changed as shown in the flow chart below:



Cruise Information Mode

The cruise information mode is displayed on area A of the multi-information display, and the display switches in the following sequence every time the DISP switch of the steering pad switch is pressed.



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Information	Outline
(1)	- Displays the value, calculated by the combination meter, which is based on the distance driven and the volume of fuel consumed (fuel injection signals from the No.1 cylinder) - Updated every 0.5 seconds.
(2)	- Displays the value, calculated by the combination meter, which is based on the distance driven since refueling and the fuel consumption volume (fuel injection signals from the No.1 injector). - The combination meter determines the vehicle has been refueled through the signal from the fuel sender gauge. - Updated every 10 seconds.
(3)	- Displays the range, calculated by the combination meter which continuously monitors and stores fuel consumption data and the residual fuel volume when IG-ON was selected. - Updated every 1 mile (U.S.A. model) or 1 km (Mexican package model).
(4)	- Displays the value, calculated by the combination meter, which is based on the distance driven since engine start. - Updated every 10 seconds.
(5)	- Displays the average speed, which is calculated by the combination meter based on the length of time and the driven distance since engine start. - Updated every 10 seconds.

Warning Mode

General

- When a warning is necessary, the warning display interrupts the multi-information display.
- The master warning light may illuminate or flash and the buzzer may sound depending on the item in the multi-information display.

Warning	Detail	Warning	Detail
Key is not detected  : Flash 080BE13Y	The key is not inside the vehicle.	Shift to IPI range  : Flash 080BE18Y	The driver door is opened with the shift lever in any position other than P.
Steering Lock is still active  : Illuminate Buzzer: Sound 080BE14Y	Steering lock has not been released.	Check Steering Lock System  : Illuminate Buzzer: Sound 080BE19Y	Steering lock is malfunctioning.
To start, depress the brake pedal, and push the engine switch Buzzer: Sound 080BE15Y	The power source is switched from OFF to ACC twice with the brake pedal released.	Check USC System  : Illuminate Buzzer: Sound 080BE20Y	A-TRC and VSC are malfunctioning.
Low Key Battery  : Flash 080BE16Y	Key battery is low.	Release the Parking Brake  : Flash 080BE21Y	Vehicle is traveling with the parking brake engaged.
 Driver's Door Open  Front Passenger's Door Open  Right Rear Door Open  Left Rear Door Open  Back Door Open  : Flash  : Illuminate 080BE17Y	Any door is open.	Moon Roof Opened  : Flash 080BE22Y	The sliding roof is open and the driver door is open.
		USC OFF  : Illuminate Buzzer: Sound 080BE23Y	VSC function is OFF

Warning	Detail	Warning	Detail
 Check Sonar  :Illuminate 080BE24Y	Clearance Sonar System is dirty or covered with ice.	 Clean Sonar  :Illuminate 080BE29Y	Clearance Sonar System is malfunctioning.
Check PCS System  :Illuminate Buzzer: Sound 080BE25Y	Pre-crash safety system is malfunctioning.	Check 4-WHEEL AHC System  :Illuminate Buzzer: Sound 081BE09Y	Active height control Suspension is malfunctioning.
PCS Not available now Buzzer: Sound 080BE30Y	Pre-crash safety system is not currently functional.	High Transmission Fluid Temperature  :Illuminate Buzzer: Sound 080BE31Y	Transmission oil temperature exceeds the applicable standards.
Low Engine Oil Level  :Illuminate Buzzer: Sound 080BE26Y	Engine oil level is low.	Check Fuel System  :Illuminate Buzzer: Sound 080BE32Y	Fuel System is malfunctioning. *1
Low Fuel  :Illuminate Buzzer: Sound 080BE27Y	Fuel level is low.	Oil Maintenance Req'd 080BE33Y	Diesel engine oil is exchange timing.*1
Fuel Filter Maintenance Req'd  :Illuminate Buzzer: Sound 081BE11Y	Fuel filter clogging is detected. *1	T-Belt Maintenance Req'd 081BE10Y	The total driving distance has reached 150,000 km or more since the last time the timing belt was replaced. *2
High Water Level  :Illuminate Buzzer: Sound 081BE12Y	Some moisture has been detected in the fuel filter. *1	-	-

*1: Diesel engine models

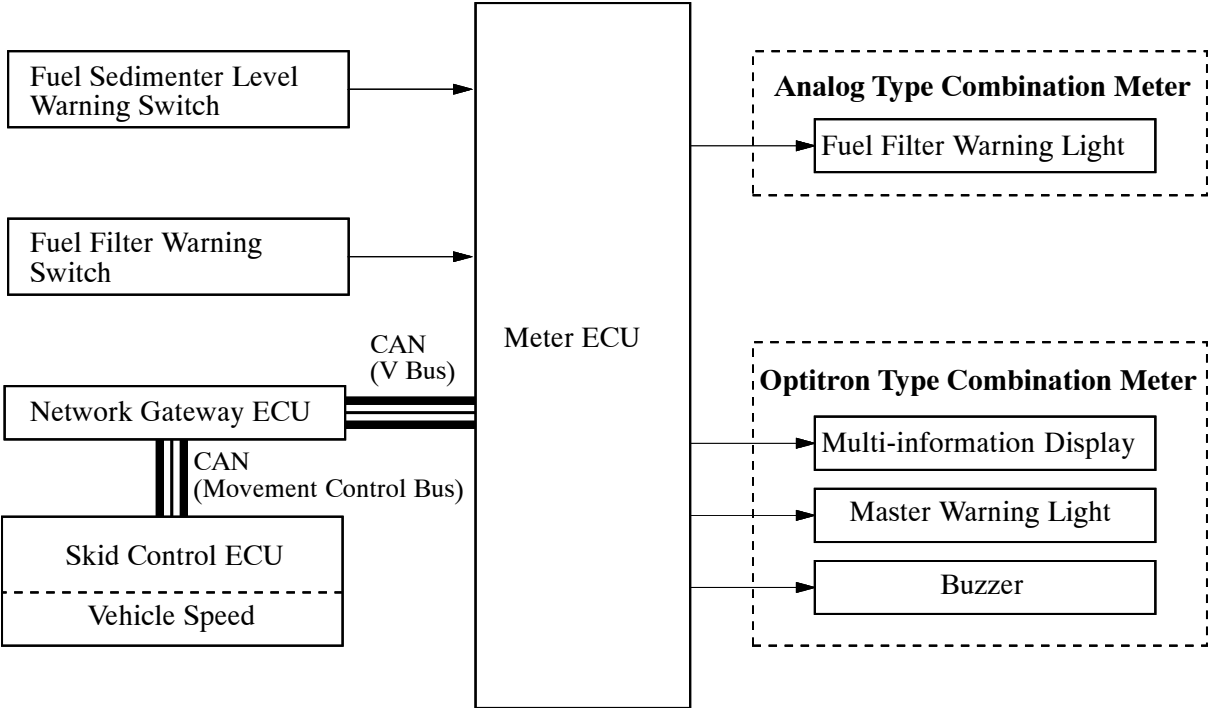
*2: 2UZ-FE engine models for G.C.C. Countries, China and general countries

5. Fuel Filter Warning (Diesel Engine Model)

General

- On the analog type combination meter, the fuel filter warning light illuminates when the fuel light is clogged or blinks when condensed water on the fuel filter is detected.
- On the optitron display type combination meter, a warning message corresponding to individual warning items is indicated on the multi-information display.
- This system is controlled by meter ECU.

► System Diagram ◀



081BE48S

► Function List ◀

Function	Analog Type Combination Meter	Optitron Type Combination Meter
Fuel Filter Warning	 Illuminate 081BE55Y	 ⚠ :Illuminate Buzzer: Sound 081BE11Y
Fuel Sedimenter Warning	 Blinking 081BE56Y	 ⚠ :Illuminate Buzzer: Sound 081BE12Y

Operation

1) Fuel Filter Warning

- The fuel filter warning switch is located in the upper area of the fuel filter.
- To prevent false alarms on a cold engine, the meter ECU starts its judgment function with the engine switch turned ON, after a total of 30 minutes have elapsed with the input of vehicle speed signal information (minimum vehicle speed of 2.5 km/h).
- The fuel filter warning switch turns OFF when the vacuum in the fuel filter reaches a predetermined level.
- When the meter ECU receives OFF signals from the fuel filter warning switch under the following conditions, it alerts the driver in accordance with the meter specifications.
 - OFF signals from the fuel filter warning switch are input continuously for 10 minutes.
 - When the engine switch is in the ON position, OFF signal (between 1 second and 10 minutes in duration) from the fuel filter warning switch is input twice with a minimum interval of 1 second.
- On all models with the optitron display type combination meter, the driver is alerted by the master warning light illuminating, a warning message appearing on the “Fuel Filter Maintenance Req’d” and a buzzer.
- On all models with the analogy type combination meter, the driver is alerted by the fuel filter warning light illuminating.

2) Fuel Sedimenter Warning

When the amount of water that has accumulated in the sedimenter reaches a predetermined level, the fuel sedimenter level switch, which is located in the bottom of the fuel filter, turns ON. The meter ECU renders this signal as a fuel sedimeter warning and then alerts the driver in accordance with the meter specifications.

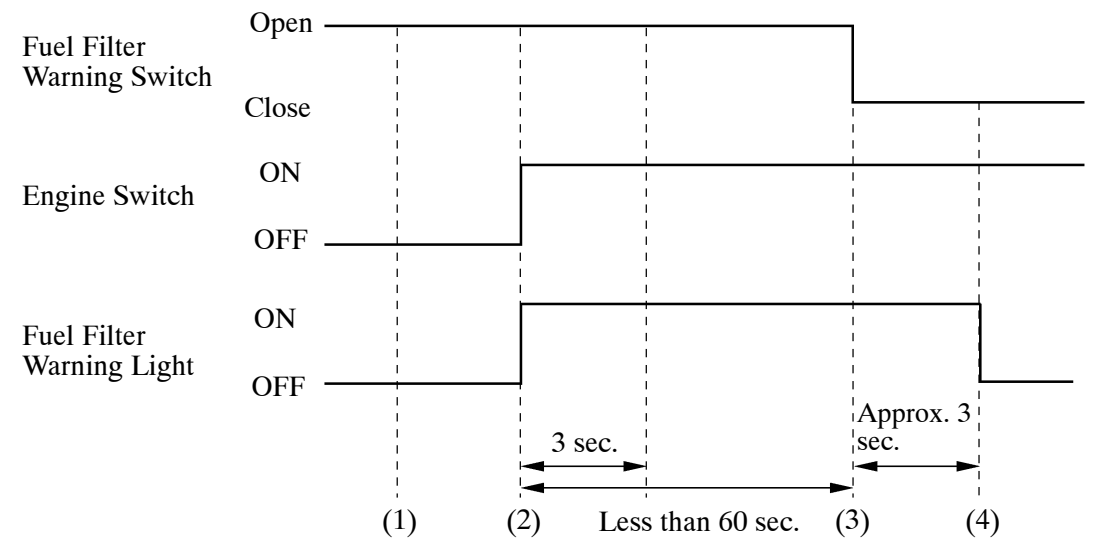
- On models with the optitron display type combination meter, the driver is alerted by the master warning light illuminating, a warning message “High Water Level” appearing on the combination meter and the buzzer sounding.
- On models with the analogy type combination meter, the driver is alerted by the fuel filter warning light blinking and the buzzer sounding.

Service Tip

When the fuel filter warning switch detects a failure, the meter ECU stores this failure in memory, and continues to illuminate* the fuel filter warning light. Therefore, after replacing the fuel filter, the illumination information stored in the meter ECU memory must be erased through the following procedure:

- (1) With the engine switch turned OFF, disconnect the fuel filter warning switch connector.
- (2) Turn the engine switch from OFF to ON.
- (3) After more than 3 seconds but less than 60 seconds have elapsed, reconnect the connector.
- (4) After approximately 3 seconds have elapsed with the fuel filter warning switch closed, the system confirms the deletion of the illumination information by turning OFF* the warning light.

*: The warning light will blink in case of a sedimenter failure.



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6. Buzzer

The table below shows the warning and reminder functions of the buzzer.

Function	Item
Warning	• Multi-information Display Warning Mode Indication (See page BE-46) • Speed Limit Warning (Only for G.C.C. Countries models) • Fuel Filter Warning (See page BE-48) • Fuel Sedimenter Warning (See page BE-48)
Reminder	• Key Reminder (See page BE-186) • Seat Belt Reminder (See page BE-229) • Oil Maintenance Reminder (See page EG-210)